

ZHYTOMYR, UkraineWMO: **333250**Lat: **50.2806N**Lon: **28.6592E**Elev: **224**StdP: **98.66**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-14.9	-20.6	0.6	-17.6	-17.8	0.8	-14.2	10.1	1.4	8.6	0.3	1.6	310	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.4	20.5	28.7	19.6	27.1	19.0	21.7	28.1	20.8	26.6	20.0	25.5	2.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.7	24.7	18.7	13.9	23.8	17.9	13.2	22.8	64.4	28.4	61.1	26.5	58.1	25.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.1	6.1	-21.7	32.7	3.9	2.0	-24.5	34.1	-26.7	35.2	-28.9	36.3	-31.8	37.8		
(5)			-21.9	23.1	3.8	1.1	-24.7	23.9	-26.9	24.6	-29.1	25.2	-31.8	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-3.8	-2.6	2.2	9.4	14.9	18.2	20.0	19.2	14.1	8.1	3.0	-1.9
	DBStd	9.58	5.70	5.71	4.61	4.20	4.14	3.38	2.98	3.29	3.88	4.34	4.71	5.49
	HDD10.0	1770	427	352	244	59	8	0	0	0	9	88	214	369
	HDD18.3	3792	685	586	501	268	122	45	17	28	136	317	461	628
	CDD10.0	1208	0	0	2	42	159	246	311	284	131	30	3	0
	CDD18.3	187	0	0	0	1	14	40	69	54	8	0	0	0
	CDH23.3	1708	0	0	0	11	147	348	591	528	82	1	0	0
	CDH26.7	434	0	0	0	2	24	68	161	164	16	0	0	0
Wind	WSAvg	2.5	2.9	3.0	2.9	2.6	2.3	2.2	2.0	1.9	2.1	2.4	2.7	2.6
Precipitation	PrecAvg	607	34	27	30	44	59	77	90	77	50	35	46	38
	PrecMax	879	93	82	69	96	115	157	207	172	185	106	124	76
	PrecMin	323	9	5	1	9	1	14	15	18	4	2	9	2
	PrecStd	108	21	17	20	24	27	36	50	43	38	25	25	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	10.4	16.4	24.5	29.2	30.3	32.5	33.1	28.7	22.6	15.9	10.4
		MCWB	5.5	7.0	9.8	15.0	18.3	20.3	21.5	21.4	18.2	15.3	12.0	8.5
	2%	DB	5.8	7.3	13.5	21.2	26.8	28.6	30.3	30.5	25.5	19.5	12.8	7.7
		MCWB	4.1	4.8	8.0	13.0	17.3	19.4	20.9	20.4	17.0	14.3	10.3	6.3
	5%	DB	4.1	5.6	11.1	18.8	24.9	27.2	28.4	28.3	23.3	16.7	11.1	6.1
		MCWB	2.9	3.9	6.7	11.6	16.7	19.1	20.3	19.4	16.2	12.9	9.3	5.0
	10%	DB	2.7	3.9	8.7	16.8	22.8	25.4	26.7	26.3	20.9	14.5	9.2	4.4
		MCWB	1.7	2.4	5.5	10.5	15.6	18.2	19.6	18.7	15.3	11.6	7.9	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.6	10.4	15.7	20.0	22.0	22.8	22.9	19.6	16.8	12.5	8.8
		MCDB	7.4	10.0	15.2	23.8	27.4	27.5	29.5	29.8	26.3	21.1	14.8	10.1
	2%	WB	4.4	5.3	8.7	13.8	18.3	20.8	21.9	21.4	18.0	14.9	10.7	6.4
		MCDB	5.5	6.9	12.4	19.2	24.6	26.5	28.6	28.0	23.5	18.3	12.5	7.4
	5%	WB	3.0	3.9	7.2	12.5	17.3	19.8	21.0	20.4	16.8	13.3	9.4	5.1
		MCDB	4.0	5.3	10.3	17.4	23.3	25.5	26.7	26.4	21.9	16.2	10.8	6.0
	10%	WB	1.8	2.6	5.7	11.3	16.3	18.9	20.2	19.4	15.7	11.8	8.0	3.5
		MCDB	2.5	3.8	8.4	15.7	21.5	24.0	25.4	24.8	20.0	14.1	9.1	4.3
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.3	6.8	9.3	10.5	10.0	10.0	10.6	9.2	7.8	4.8	4.2
		MCDBR	4.4	6.5	10.7	12.7	13.1	12.5	12.8	13.8	13.2	11.2	7.6	5.1
		MCWBR	3.6	4.8	6.3	6.1	5.6	5.3	5.1	5.3	6.0	6.4	5.3	4.2
	5% WB	MCDBR	4.1	6.0	9.4	11.1	11.9	11.0	11.2	12.1	11.6	10.2	6.5	4.7
		MCWBR	3.6	4.7	6.1	5.9	5.6	5.4	5.1	5.3	5.8	6.2	4.8	4.1
Clear-Sky Solar Irradiance	taub	0.309	0.339	0.376	0.431	0.405	0.401	0.430	0.431	0.397	0.360	0.341	0.307	
	taud	2.207	2.161	2.187	2.124	2.275	2.306	2.235	2.227	2.280	2.355	2.302	2.259	
	Ebn at Noon	702	774	816	809	852	857	823	801	788	751	662	647	
	Edh at Noon	75	101	118	141	127	124	131	126	108	84	69	62	
All-Sky Solar Radiation	RadAvg	0.87	1.67	2.94	4.34	5.59	5.92	5.65	4.97	3.42	2.01	0.82	0.60	
	RadStd	0.15	0.32	0.47	0.46	0.48	0.43	0.39	0.28	0.53	0.32	0.10	0.14	